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SMARTH2WO: IOT-BASED SMART WATER DISPENSER SYSTEM WITH PREDICTIVE MAINTENANCE USING MACHINE LEARNING

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CHAPTER 1:

INTRODUCTION / BACKGROUND OF THE STUDY

Access to clean and safe drinking water is a basic necessity that directly affects the health, focus, and productivity of students and personnel in educational institutions. However, many schools continue to experience difficulties in maintaining a consistent and accessible source of potable water. At the Institute of Technology (ITECH) of the Polytechnic University of the Philippines, several students report inconvenience due to malfunctioning water dispensers and the absence of reliable facilities in high-traffic areas. Most existing units rely on an honesty-based payment setup, which often leads to inconsistent collection and limited accountability. As a result, students frequently experience difficulty accessing drinking water during long lectures or laboratory sessions.

The problem is further aggravated by the growing use of single-use plastic bottles, which contribute to increasing plastic waste on campus and in nearby



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communities. Several studies have shown that water refill stations can significantly reduce plastic bottle consumption while encouraging environmentally responsible habits among students (Piccirillo-Stosser, 2018; Uenaka & Yasui, 2018). Soubra et al. (2025) also noted that awareness of environmental impact and ease of access both influence students' willingness to use refillable alternatives. These findings highlight the importance of providing convenient and modernized solutions that promote sustainable consumption behavior within the university setting.

The advancement of Internet of Things (IoT) and Machine Learning (ML) technologies also presents new opportunities for system monitoring and predictive maintenance. Recent studies have demonstrated the use of sensors, flow meters, and microcontrollers to achieve precise water dispensing and real-time tracking of system performance (Amarul & Ling, 2025; Manjulamma et al., 2024). Additionally, predictive models such as Random Forest and Gradient Boosting have been applied successfully in analyzing water usage data and detecting operational anomalies (Görenekli & Gülbağ, 2024; Robles-Velasco et al., 2021). Through these methods, administrators can anticipate maintenance needs, prevent dispenser malfunctions, and ensure continuous service availability.

In response to these challenges, this study proposes the development of SmarH2WO: An IoT-Based Smart Water Dispensing System with Predictive Maintenance Using Machine Learning. The system seeks to replace unreliable honesty-based setups with an automated coin-operated mechanism integrated with sensor-based volume control and real-time IoT monitoring. By incorporating a predictive maintenance layer, the system can anticipate refilling requirements and potential component failures, ensuring continuous service and a reliable hydration source for the ITECH community.



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THEORETICAL FRAMEWORK

This section offers numerous explanations to support the researcher's findings. Several theories and models support the development and functionality of an IoT-based automated system with predictive maintenance capabilities.

1. Internet of Things (IoT) Theory

The IoT refers to the interconnection of devices through the internet, enabling data collection, communication, and automation. This study applies IoT principles to the Internet of Things (IoT), which refers to the interconnectedness of devices via the internet, which facilitates data collection, communication, and automation. This study applies IoT principles to enable real-time monitoring of the water vending system. It tracks water levels, usage patterns, and system status to ensure efficient operation and maintenance. Allow real-time monitoring of the water vending system, tracking water levels, usage patterns, and system status to ensure efficient operation and maintenance. (Reference: Ashton, K. (2009). "That 'Internet of Things' Thing." RFID Journal.)

2. Machine Learning and Predictive Maintenance Model

Predictive maintenance theory employs machine learning to analyze historical data and predict equipment failures or resource depletion before they occur. This study incorporates this concept to forecast when water refills are needed or when components may malfunction, ensuring continuous operation and minimal downtime. (Reference: Mobley, R. K. (2002). An Introduction to Predictive Maintenance."

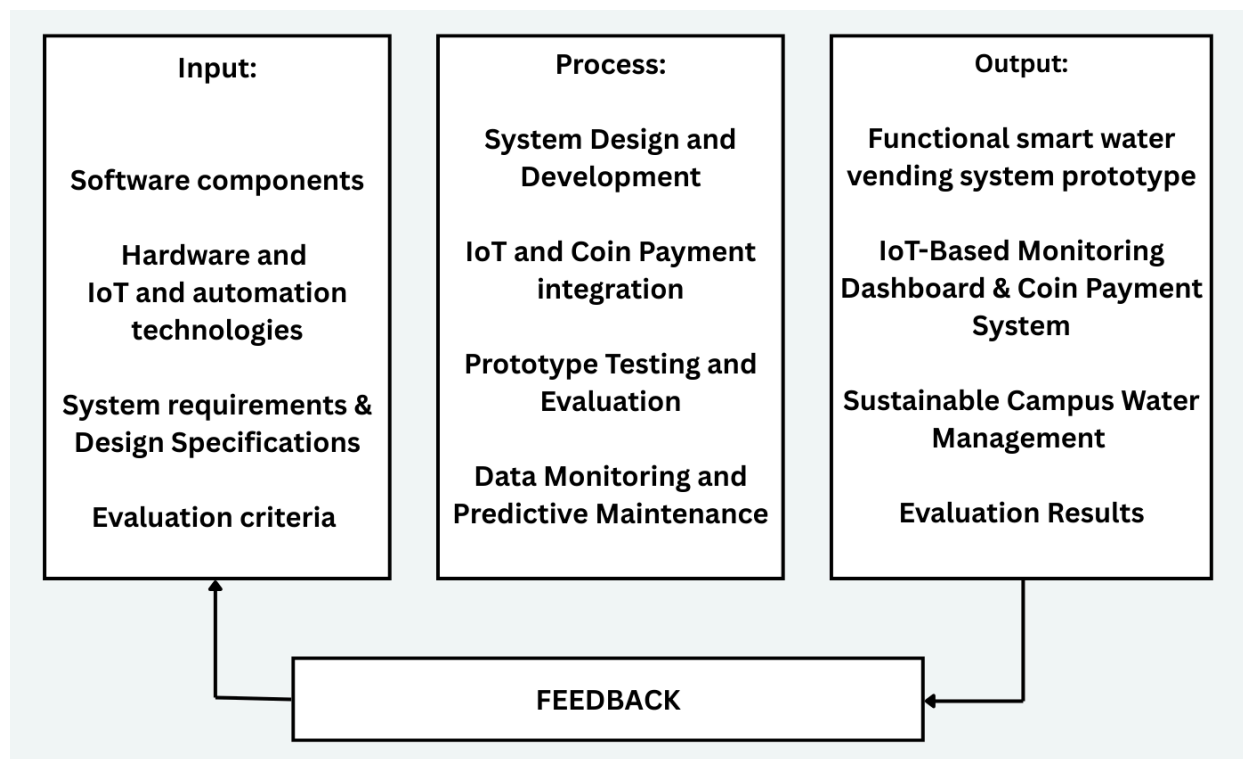
3. Automation and control system theory



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Rooted in control engineering, this theory emphasizes the use of sensors, actuators, and feedback mechanisms to achieve desired system outputs. In this study, sensors (flow, ultrasonic) and microcontrollers automate water dispensing and ensure accuracy and reliability during operation. (Reference: Nise, N.S.(2011). Control System Engineering.)

CONCEPTUAL FRAMEWORK



The study was guided by a paradigm utilizing a system approach involving three frames. The page shows graphically the model that serves as a guide for the researcher in the development of the Refillable Water Dispensing Station with Coin Payment.

The first frame shows the input phase, which included the software components, Hardware and IOT and automation technologies, System requirements, & Design Specification. It also covers the evaluation criteria that determine the performance and usability of the prototype.



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The process phase, which includes System Design and Development, IOT and coin payment integration, prototype testing and evaluation, data monitoring, and machine learning-based predictive maintenance, is depicted in the second frame.

The third frame represents the output phase, which presents the functional prototype of the Smart Water Vending System, the IoT-based monitoring dashboard with Coin payment, sustainable campus water management practices, and the evaluation results of the system's performance

STATEMENT OF THE PROBLEM

Access to a reliable and convenient source of drinking water remains a concern within the Institute of Technology (ITECH) of the Polytechnic University of the Philippines. Existing water dispensers commonly rely on honesty-based payment systems and manual monitoring.

In addition, the absence of system monitoring makes it difficult for administrators to track water usage, detect operational issues, and plan maintenance activities efficiently. Without usage data and system status information, maintenance is often reactive rather than preventive, leading to dispenser malfunctions and service interruptions.

In response to these concerns, this study seeks to address the following research problems:

1. What is the level of acceptability of the water dispenser system as evaluated by the respondents in terms of:
 - 1.1. System Reliability
 - 1.2. Dispensing Accuracy, and
 - 1.3. Ease of use for students and staff
2. How can IoT-based monitoring be implemented in the proposed water dispensing system to record and display:
 - 2.1. Water usage data,



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- 2.2. Transaction activity, and
- 2.3. Basic system status Information
3. How can the collected usage and system data support basic maintenance planning and improve the overall reliability of the water dispensing system through machine learning?

OBJECTIVE OF THE STUDY

1. System Design and Prototype Development Objectives

- To identify and address the technical challenges involved in designing a coin-operated water dispensing system that ensures accurate volume control and reliable operation.
- To develop a microcontroller-based system that integrates sensors, actuators, and control logic for automated water dispensing based on user-selected volumes.
- To assess the functionality, accuracy, and reliability of the developed prototype through controlled testing of its dispensing and payment mechanisms.

2. System Monitoring and Data Management Objectives

- To implement an IoT-enabled monitoring system that records water usage, transaction data, and system status using a centralized database.
- To develop a dashboard that allows administrators to view system metrics such as water dispensed, number of transactions, and operational status.

3. Data Analysis and Maintenance Support Objectives

- To analyze collected system data in order to identify usage patterns and potential indicators of maintenance requirements.
- To apply machine learning techniques for predicting maintenance needs and supporting informed system management decisions.



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SCOPE AND DELIMITATION

This study focuses on the design, development, and partial implementation of a coin-operated refillable water dispensing system intended for use within the Institute of Technology (ITECH) building of the Polytechnic University of the Philippines – Sta. Mesa.

The scope of the study includes the development of a functional prototype that automates water dispensing through user-selected volume options and coin-based payment using a universal coin acceptor. The system integrates sensors and microcontroller-based control logic to ensure accurate dispensing and reliable operation. An IoT-enabled monitoring component is included to record water usage, transaction data, and basic system status information, which will be displayed through a web-based dashboard.

The study also covers the preliminary application of machine learning techniques for basic predictive maintenance and refill planning. Machine learning implementation in this phase is limited to analyzing historical usage and system data to support maintenance-related decision-making and does not include real-time model deployment or advanced optimization.

The study is delimited to coin-based transactions only. Cashless and online payment methods such as QR codes, e-wallets, and mobile applications are excluded from the current implementation. The system does not include water purification, filtration, or treatment processes, as it assumes the use of an existing potable water source.

Furthermore, the prototype will be developed and evaluated as a single-unit system within the ITECH building. Expansion to multiple units, multi-floor deployment, or integration across other PUP campuses is beyond the scope of this study and is



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reserved for future work. The functionality of IoT features depends on the availability of a local Wi-Fi network, and system performance under unstable or unavailable network conditions is not evaluated.

KEYWORDS: Water Dispenser, IoT, Predictive Maintenance, Coin-based Payment, Machine Learning

SIGNIFICANCE OF THE STUDY

This proposed project is significant because of the benefit it will bring to the following groups of persons:

ITECH Students & Faculty. This study will ease the burden of finding drinking water by providing a reliable, accessible, and safe source within the ITECH building. Utilizing an automated coin-operated system ensures that users can obtain water efficiently without relying on manual transactions or honesty-based dispensers.

Broader Institutional Impact. Administrators can schedule maintenance, save expenses and waste, and support PUP's environmental objectives with real-time dashboards. According to the theory, this improves user convenience over trust-based systems and shows how technology is adopted in poor nations like the Philippines.

Future Researchers. This study can benefit future researchers as it provides an essential baseline for conducting an offline, coin-operated mechanism paired with predictive maintenance logic. This research offers a technical framework for creating sustainable campus infrastructure.



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DEFINITION OF TERMS

Coin-Based Payment System – Refers to the automated mechanism using a universal coin acceptor that triggers the water dispensing process upon detecting the required Philippine peso denominations.

Flow Sensor – A hardware component used to measure the volume of water passing through the system to ensure the accurate delivery of the user's selected amount (e.g., 250ml or 500ml).

Internet of Things (IoT) – The network of physical objects (sensors and microcontrollers) in the dispenser that collect and exchange data with a web-based dashboard for real-time monitoring.

ITECH Community – Refers to the students, faculty, and administrative staff of the Institute of Technology at the Polytechnic University of the Philippines – Manila, who serve as the primary end-users of the system.

Machine Learning (ML) – In this study, it refers to the use of algorithms (such as Random Forest or Gradient Boosting) to analyze historical usage data to predict when the water supply needs refilling or when parts require maintenance.

Microcontroller – The "brain" of the system (likely an Arduino or ESP32) that processes inputs from the coin acceptor and sensors to control the solenoid valve and upload data to the cloud.

Potable Water – Water that is safe for human consumption, which in this study is assumed to be provided by the existing university supply or pre-purified containers.

Predictive Maintenance – A proactive maintenance strategy that uses data patterns to anticipate equipment failure or service needs before they occur, thereby reducing system downtime.



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Solenoid Valve – An electromechanical device used to control the flow of water; it opens when a valid payment is received and closes once the target volume is reached.

System Reliability – The ability of the Smarth2WO prototype to perform its required functions under stated conditions for a specified period without failure.

Universal Coin Acceptor – An electronic device programmed to recognize and validate different denominations of coins (e.g., 1, 5, and 10 Peso coins) to prevent the use of tokens or slugs.

Web-based Dashboard – The graphical user interface where administrators can remotely view system metrics such as total water dispensed, transaction history, and operational status.



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CHAPTER 2:

REVIEW OF RELATED LITERATURES

This chapter examines a wide range of studies that define the current technological and environmental state of automated water systems. As colleges transition toward "smart campus" models, combining IoT connectivity with Machine Learning has become a practical way to ensure these systems actually work when students need them. This review also looks at the environmental side—specifically how these stations help cut down on plastic bottle waste and encourage better hydration habits on campus. By evaluating existing hardware and what users actually think of it, this section provides the groundwork for building a more reliable, refillable water station specifically for the ITECH community.

THEMATIC ARRANGEMENT OF ARTICLES

AUTOMATED WATER DISPENSING SYSTEMS AND INTEGRATION

Recent studies emphasize the development of automated water dispensing systems that integrate microcontrollers, sensors, and IoT technologies to enhance precision, efficiency, and user convenience. Amarul and Ling (2025) designed an Arduino Uno-based system capable of dispensing preset volumes (100 mL, 250 mL, 500 mL) with 99.8% accuracy, reducing water wastage by 30% and energy consumption by 20%. Koh et al. (2024) implemented a Smart Water ATM using Arduino Mega and Uno boards, incorporating IR sensors, flow sensors, and solenoid valves to maintain precise dispensing while preventing spillage. Manjulamma et al. (2024) demonstrated the value of IoT-enabled monitoring through ultrasonic sensors and GSM/WiFi modules, supporting proactive maintenance and minimizing downtime. Gagana et al. (2023) further reinforced hygienic dispensing through IR sensors, highlighting the importance of non-contact water delivery in institutional settings.



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Collectively, these studies form a technological foundation for the proposed refillable water dispensing station, demonstrating feasibility, scalability, and operational reliability in high-traffic academic environments.

WATER ATMS AND REFILL STATION IMPLEMENTATIONS

Commercial and pre-built water ATMs have become increasingly common in public spaces, providing automated dispensing and cashless payment features. Studies such as Gupta (2022) and Badodkar et al. (2024) show that these systems successfully integrate user authentication, volume selection, and payment mechanisms, offering convenience and reducing manual oversight. However, many implementations remain limited in adaptability to campus environments, lack predictive maintenance, or are cost-prohibitive for small-scale deployment. The proposed study differentiates itself by combining IoT-enabled monitoring and machine learning-based predictive maintenance, creating a solution tailored for educational institutions that addresses reliability, sustainability, and scalability (Manjulamma et al., 2024; Koh et al., 2024).

COIN-BASED PAYMENT SYSTEMS

Coin-operated water dispensers have historically provided a simple, cost-effective method for regulating access and maintaining hygiene in public and institutional contexts. Gupta (2022) highlighted precise volume delivery per coin using microcontroller-based systems, while Manjulamma et al. (2024) observed their efficiency in reducing queues and promoting consistent water availability.

MACHINE LEARNING FOR PREDICTIVE MAINTENANCE

Predictive maintenance using machine learning (ML) has been successfully applied to water networks and dispenser systems to forecast equipment failures and optimize operational scheduling. Robles-Velasco et al. (2021) reviewed regression, support vector regression (SVR), and artificial neural networks (ANNs) for predicting remaining useful life and failure timing. Austria and Lacbay (2025) applied similar



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approaches in university contexts to forecast water usage and optimize refill schedules. Comparative studies (Görenekli & Gülbağ, 2024; Konate et al., 2024; Obinwanne et al., 2023) evaluated algorithms including Random Forest (RF), Gradient Boosting Machines (GBM), and LSTM networks, demonstrating that ensemble methods offer superior predictive accuracy for usage and anomaly detection. Integrating these models with IoT-enabled sensors allows data-driven decision-making, ensuring uninterrupted service and efficient resource allocation in high-traffic campus environments. Reliability is often the biggest challenge for school-based systems. Jundun et al. (2023) addressed this by using a Random Forest algorithm to turn simple sensor data into smart predictions. Instead of waiting for the water to run out or the machine to break, their model forecasts exactly when maintenance is needed.

SUSTAINABILITY AND PLASTIC CONSUMPTION BEHAVIOR

Environmental concerns over single-use plastic bottles have led to the adoption of Water Refill Stations (WRS) as sustainable alternatives in educational institutions. Piccirillo-Stosser (2018) reported that refill stations could reduce single-use plastic consumption by millions of bottles annually, while Uenaka and Yasui (2018) highlighted their contribution to campus sustainability and eco-friendly hydration behaviors. Behavioral research indicates that subjective norms, perceived convenience, and environmental awareness strongly influence students' willingness to use refill stations (Soubra et al., 2025). Providing accessible, reliable, and sustainable water dispensing systems encourages responsible consumption, complementing the technical and operational benefits of automated water dispensers.

INSTITUTIONAL ADOPTION AND USER EXPERIENCE OF SMART WATER SYSTEMS

Beyond technical performance, user experience is crucial for the successful adoption of smart dispensers. Koh et al. (2024) and Badodkar et al. (2024) emphasize that intuitive interfaces, remote monitoring, and dynamic volume control maximize user



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satisfaction. Ease of use, speed, and reliability are consistently cited as critical factors for institutional acceptance. These insights underscore that the effectiveness of a smart water dispensing prototype depends not only on hardware and software integration but also on human-centered design to ensure long-term adoption and compliance.

SYNTHESIS OF RELATED STUDIES

The research surrounding water delivery systems has clearly moved past simple mechanical designs toward a more "intelligent" approach that utilizes the Internet of Things (IoT) for predictive maintenance and sustainability. A recurring theme in the literature suggests that while contemporary campuses still find value in the cost-effective nature of coin-operated systems, these fundamental designs must now be augmented with modern connectivity to meet current institutional standards. The work of Amarul and Ling (2025) and Koh et al. (2024) specifically highlights that microcontroller-based systems are no longer just about convenience; they are about achieving a level of precision through flow sensors and solenoid valves that manual systems simply cannot match.

In the context of the ITECH campus, the integration of machine learning provides a much-needed layer of dependability. By moving away from a "fix-it-when-it-breaks" mindset, researchers are now prioritizing foresight. According to Jundun et al. (2023), keeping a system running smoothly requires more than just good hardware; by applying a Random Forest model to IoT data, they showed that machines could essentially 'predict' their own maintenance needs. This supports the integration of predictive models—such as the Random Forest and LSTM algorithms discussed by Robles-Velasco et al. (2021) and Austria and Lacbay (2025)—to allow for a proactive maintenance schedule that keeps the system online during peak hours. Using data to prevent downtime in this way ensures that the dispenser remains a dependable resource for the campus community rather than a maintenance burden.



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Lastly, the body of research indicates that the success of a campus water station depends on the balance between backend precision and human-centered design. While the hardware ensures accuracy, factors like user interface, transaction speed, and operational uptime are what determine institutional acceptance. As noted by Piccirillo-Stosser (2018) and Soubra et al. (2025), a refill station only effectively reduces plastic waste if it is perceived as a reliable, high-tech alternative that is just as easy to use as buying a bottled drink. By synthesizing these technical and behavioral insights, the proposed station at ITECH aims to bridge the gap between industrial-grade water ATMs and the specific sustainability needs of a university setting, creating a scalable solution that improves the daily experience of students and staff alike.



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CHAPTER 3:

METHODOLOGY

The methodical approach used for preparing, constructing, and assessing the Refillable Water Dispensing Station with Coin Payment is described in this chapter. A systematic methodological framework was used at the Polytechnic University of the Philippines-Institute of Technology (PUP-ITECH) to address the issues of water accessibility and reliable payment systems. The Descriptive-Developmental study design, the prototype's developmental life cycle, participant selection, and data collection tools are all described in this section. It also describes the ethical and analytical methods used in ensuring the suggested system's technical reliability and user-based viability.

RESEARCH DESIGN

This study will use a Descriptive-Developmental Research Design, which combines both descriptive and developmental approaches to identify existing problems, create a practical solution, and evaluate its expected usability. The descriptive part of the study will focus on the current issues concerning drinking water accessibility, the inefficiency of trust-based payment systems, and the growing problem of plastic waste within the Institute of Technology (ITECH) at the Polytechnic University of the Philippines. Meanwhile, the developmental part will focus on designing, building, and testing a working prototype of a Refillable Water Dispensing Station with Coin Payment.

DATA GENERATION PROCEDURE

In the first phase, we will conduct a preliminary observation of the PUP-ITECH building's current water dispenser. This stage will enable us to pinpoint common problems, constraints, and opportunities for enhancement in the existing water



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dispensing system. The proposed Water Dispenser will be designed using the collected data.

The second phase involves the design and development of the coin-operated water dispensing prototype. This includes the integration of the universal coin acceptor, sensors, control logic, and microcontroller-based automation to ensure accurate water dispensing and reliable system operation.

The final prototype will be tested experimentally in the third phase. We will evaluate the dispenser's functionality, effectiveness, transaction success rate, and dependability under controlled conditions throughout this phase. For future enhancement, we will monitor metrics like response time, payment success rate, and operational performance in addition to any technological issues that may come up.

Lastly, a survey will be administered in the fourth phase to get feedback from selected PUP-ITECH community members, including students, instructors, and staff. User satisfaction, perceived convenience, ease of use, and desire to adopt the system will all be evaluated by the survey questionnaire. The goals of the study will be explained to participants, and their responses will be kept completely private.

After completing all four processes, the acquired data will be compiled, organized, and tabulated for analysis. This systematic process will provide both technical and user-based insights for assessing the overall effectiveness of the proposed Water Dispenser system.

DATA CASE ANALYSIS

Descriptive statistical methods will be used to analyze the data collected from system testing and survey questionnaires. Frequency counts, percentages, weighted means, and ranking will be applied to evaluate the level of acceptability of the system in terms of reliability, dispensing accuracy, and ease of use.



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System performance data such as dispensing accuracy, transaction success rate, and response time will be summarized using descriptive metrics to assess the operational reliability of the prototype.

For the machine learning component, historical water usage and transaction data collected from the system will be analyzed to identify usage patterns and potential indicators of maintenance requirements. The machine learning analysis in this phase is limited to exploratory and predictive modeling for maintenance planning purposes and will not involve real-time deployment.

RESEARCH LOCALE, POPULATION, SAMPLING

Locale of the Study

The study will be conducted at the Polytechnic University of the Philippines (PUP), specifically within the Institute of Technology (ITECH) campus located in Pureza, Sta. Mesa, Manila. Established as a premier polytechnic institution, PUP provides a high-traffic academic environment that is ideal for testing the efficiency of automated systems. The researchers chose this place of implementation because the campus's large student population and existing infrastructure offer a plenitude of information necessary to evaluate the scalability and operational reliability of an IoT-based water dispensing station.

Population of the Study

The target population of this study includes the students and staff members of the ITECH community who are the primary users of the current campus water dispenser. Specifically, the student population is drawn from the **Diploma in Computer Engineering Technology (DCPET)** program, focusing on the following third-year sections:



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- DCPET 3-1
- DCPET 3-2
- DCPET 3-3

The researchers specifically targeted the third-year students of the Diploma in Computer Engineering Technology (DCPET) program, focusing on sections 3-1, 3-2, and 3-3. These sections were chosen because their technical curriculum provides them with the background knowledge necessary to evaluate the hardware and software components of the prototype effectively.

Sampling

The researchers will utilize the Purposive Sampling technique to select a total of 75 respondents. This method ensures that the participants are chosen based on their regular use of the current water dispenser in ITECH and their technical ability to provide constructive feedback during the testing phase.

The sample size is distributed as follows to ensure a balanced evaluation:

- **60 Student Respondents:** 20 representatives from each of the three DCPET sections mentioned above.
- **15 Staff Respondents:** Selected from various administrative and laboratory offices within the ITECH building.

By choosing 75 respondents, the researchers aim to reach a data saturation point where common technical issues and user preferences can be clearly identified without overextending the logistical limits of testing a single hardware prototype.

STATISTICAL TREATMENT AND MACHINE LEARNING

The researchers will utilize **Descriptive Statistics** to evaluate the data gathered from the 60 students and 15 staff members. This approach allows the study to transform



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raw survey responses into measurable insights regarding the prototype's performance and general user satisfaction.

Weighted Mean. This will be used to represent a numerical average response for each item on the Likert Scale (e.g., 1–5 rating for "Ease of Use").

$$\bar{x} = \frac{\sum w_i x_i}{\sum w_i}$$

Percentage Distribution. This will be used to describe the demographic distribution of the respondents, such as the ratio of students to staff. The frequency in the category is divided by the total number of respondents and then multiplied by 100% to get the percentage.

$$\% = \frac{f}{N} \times 100$$

Likert Scale. A 5-point scale will be used to categorize the responses, ranging from "Strongly Agree" to "Strongly Disagree."

Mean Absolute Error (MAE). To measure the gap between the intended volume (e.g., 100mL) and the actual volume dispensed by the flow sensors, Mean Absolute Error (MAE) will be utilized.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Transaction Success Rate (TSR). The researchers will utilize this tool to further validate the prototype's operational stability. This metric is essential for a coin-operated system because it measures how often the machine completes a cycle (payment to dispensing) without error.



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$$TSR = \left(\frac{\text{Successful Transactions}}{\text{Total Transaction Attempts}} \right) \times 100$$

Random Forest (RS). This machine learning algorithm will be utilized to analyze historical sensor data to predict equipment failures and resource depletion. Random Forest operates by constructing a multitude of decision trees during training and outputting the average prediction of the individual trees.

$$RF = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

ETHICAL CONSIDERATIONS

The researchers shall closely abide by the academic rules and ethical standards set forth by the Polytechnic University of the Philippines. This study prioritizes the informed consent and voluntariness of all participants, ensuring that ITECH students and staff are thoroughly briefed on the study's aims before interacting with the prototype or completing the survey. In full accordance with the Data Privacy Act of 2012 (R.A. 10173), the IoT-enabled monitoring system is intended for anonymity, ensuring that no personally identifying information is gathered, kept or compromised. By using food-grade materials and appropriately insulating electronic components from the water supply to prevent risks, the researchers uphold the idea of non-maleficence with regard to the physical prototype. Additionally, by properly citing all secondary sources and publishing all sensor data and machine learning results in an honest and upfront manner, the study upholds the highest standards of academic integrity. The manuscript will go through extensive plagiarism testing in compliance with the university's intellectual property rules to guarantee that the work is entirely original.

MATERIAL INSTRUMENTS

This study utilizes two primary research instruments: a system performance testing setup and a survey questionnaire.

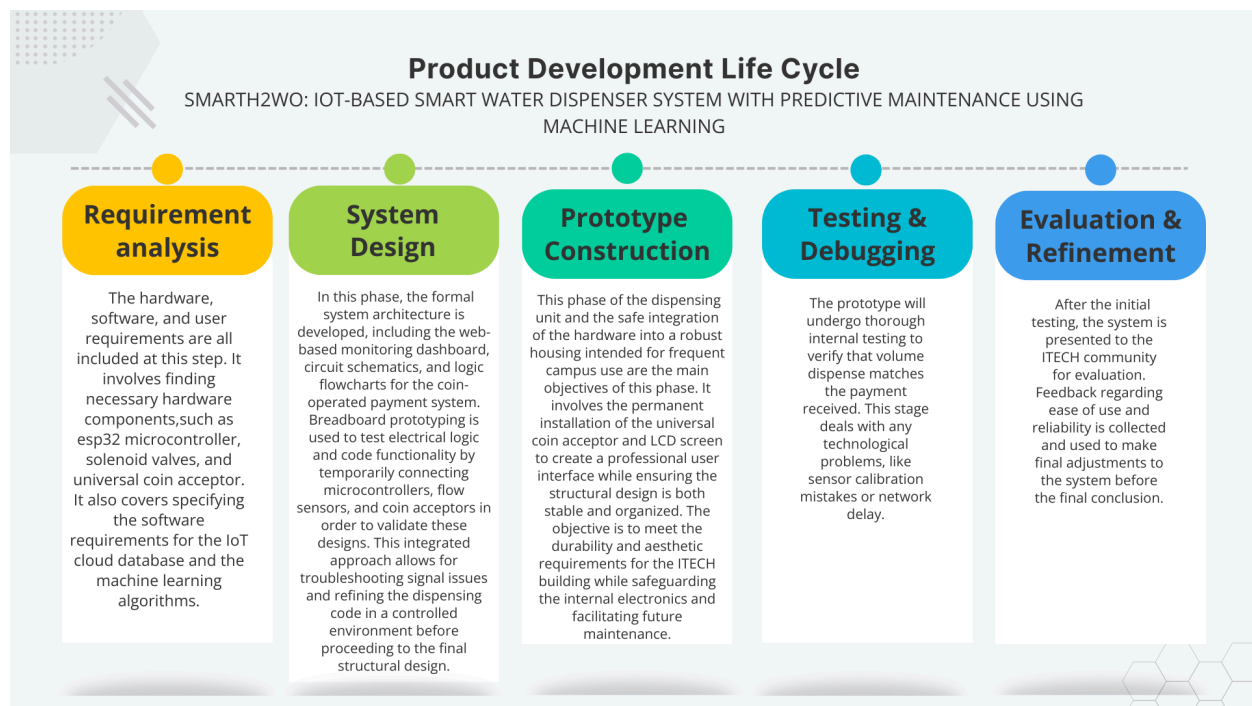


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The system performance testing setup consists of the developed coin-operated water dispensing prototype. This instrument is used to evaluate the technical performance of the system in terms of dispensing accuracy, transaction success rate, response time, and operational reliability. Controlled testing will be conducted to verify that the system dispenses the correct water volume based on user selection and successfully activates only after valid coin insertion.

The survey questionnaire will be used to measure the level of acceptability of the proposed system among selected students, faculty members, and staff of the ITECH building. The questionnaire is designed to assess user perceptions in terms of system reliability, ease of use, and dispensing accuracy. Responses will be measured using a 5-point Likert scale ranging from Strongly Disagree to Strongly Agree.

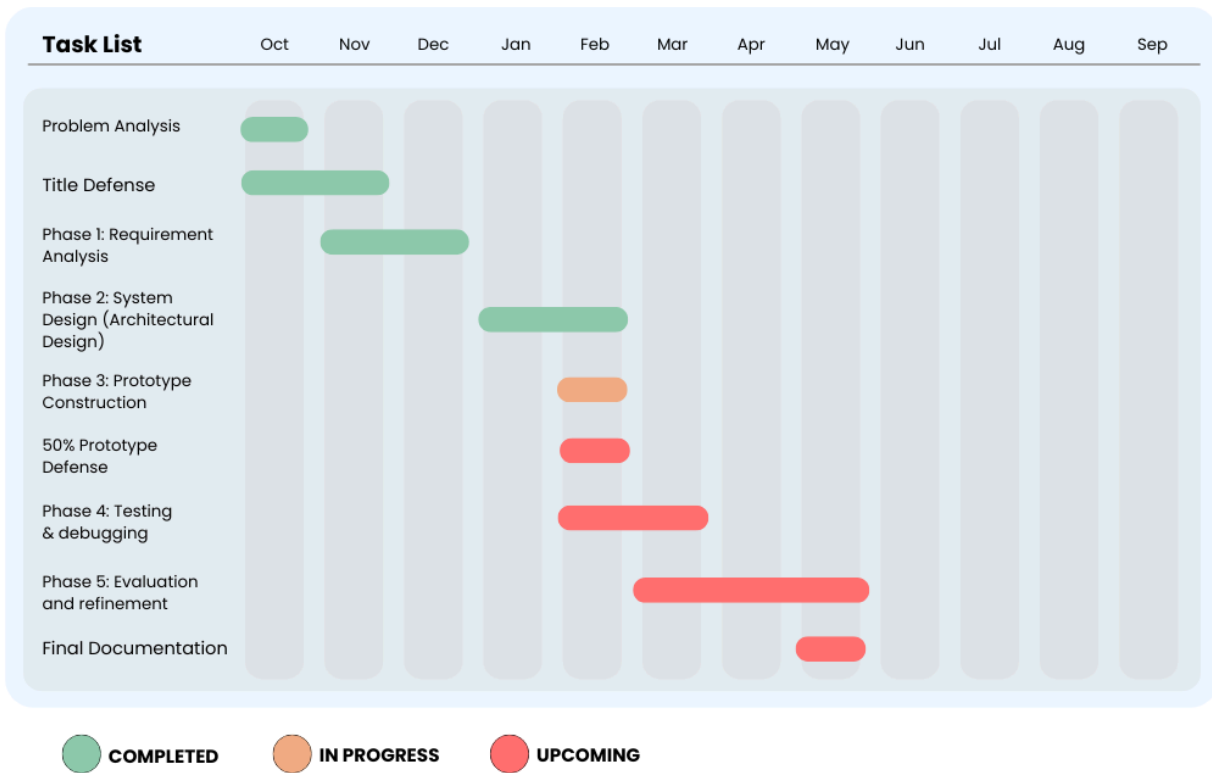
PROTOTYPE DEVELOPMENT LIFE CYCLE



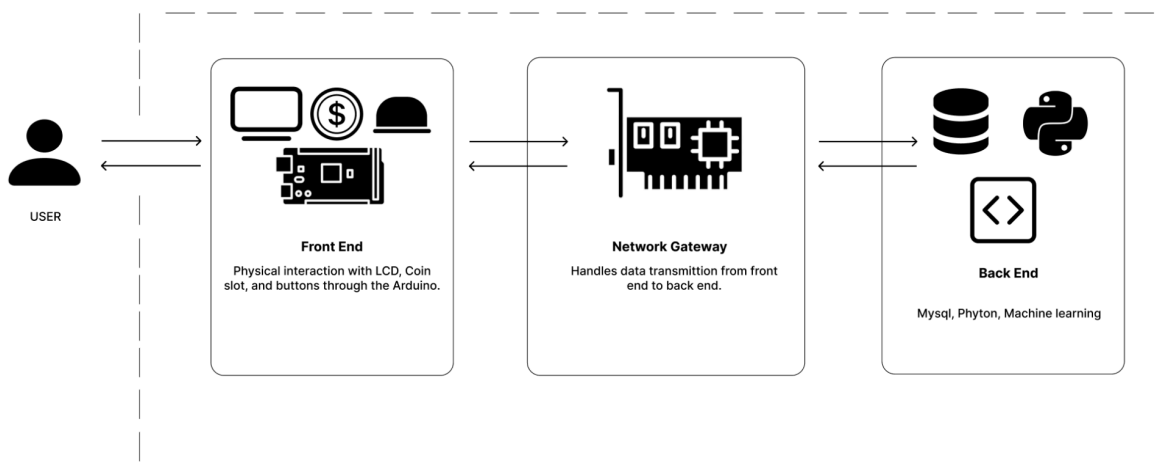


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TIME FRAME CHART



PROPOSED PROTOTYPE ARCHITECTURE





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